

PARTS LIST

Semiconductors:

IC1	- 7812, 12V regulator
IC2	- CD4060 oscillator and counter
T1	- BC327 pnp transistor
T2, T4	- BC337 npn transistor
T3, T5-T12, T14	- BC547 npn transistor
T13	- BC557 pnp transistor
SCR1, SCR2	- BT169 silicon control rectifier
D1-D9	- 1N4007 rectifier diode
D10, D11	- 1N4148 switching diode
LED1-LED4	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon)

R1-R3, R5, R10, R20, R23-R25, R29, R32, R34, R36, R37	- 1-kilo-ohm
R4, R6, R8, R9, R12, R13, R16, R18, R19, R21, R31, R33, R35	- 10-kilo-ohm
R11, R14, R15, R17	- 4.7-kilo-ohm
R7, R26	- 47-kilo-ohm
R22, R27, R30	- 100-kilo-ohm
R28	- 1 mega-ohm

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 10 μ F, 35V electrolytic
C3, C5	- 0.1 μ F ceramic disk
C4	- 220 μ F, 35V electrolytic
C6-C9	- 10 μ F, 35V electrolytic
C10	- 0.47 μ F ceramic disk

Miscellaneous:

X1	- 230V AC primary to 12V, 1A secondary transformer
RL1	- 12V, 1C/O relay
RL2	- 12V, 2C/O relay
S1	- Push-to-off tactile switch
S2, S3	- DIP switch (4-way)
S4-S6	- On/off switch

supply is given to the circuit, resistors R6 and R12 provide positive bias to transistor T4. Transistor T4 conducts and relay RL2 gets energised. The N/O contacts of relay RL2 close to activate the pump starter and the pump starts. Capacitor C4 starts charging through R7. Sufficient voltage is built across capacitor C4 within two to four seconds to saturate transistor T6, resulting in cutting off transistor T4 to de-energise relay RL2.

Power supply

The 12V, 500mA regulated power supply is obtained from mains, step-down transformer and a 12V voltage regulator circuit. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 12V. The transformer output is rectified by

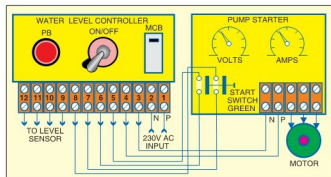


Fig. 5: Connection diagram of the automatic water level controller

a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7812 (IC1). Capacitors C2 and C3 bypass the ripples present in the supply. LED1 acts as the power indicator and resistor R1 limits the current through LED1.

PCB and component layout

An actual-size, single-side PCB for the circuit of automatic water level controller is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on a PCB to minimise assembly time and errors. Use proper heat sink for voltage regulator IC.

Once the PCB assembly is completed it can be housed in a suitable cabinet and connected to other components with wires of different colours. Use heavy-duty insulated copper wire for wiring relays RL1, RL2, and the terminal connectors to handle about 20A (for 1 HP motor). Fix on/off switch S6, reset switch S1 and LEDs on the cabinet cover.

After completing the wiring the controller box can be tested for its proper functioning before connecting to the pumps. It is suggested to fabricate the level sensor probes using stainless steel screws to avoid rusting.

Testing of the controller unit

Testing procedure for the controller unit is given below:

1. Connect level sensors A, B and C, and short A' and D with the help of a jumper wire.

2. Connect a 60W bulb (in place of motor) at output points 3 and 4 of the terminal connectors as shown in Fig. 5.

3. Close the on/off switch S6 and keep MCB in 'on' position.

4. Connect 230V AC input at terminal 1 and 2.

5. As the sensors are not in contact with water, 'pump on' indication is shown by glowing of the green LED1. Relay RL1 energising is indicated by glowing of the bulb.

6. Dip sensors A and then B in a mug of water. The green LED and the bulb must remain on.

7. Dip sensor C in water. The green LED and bulb must go off.

8. Remove sensor C from water. The green LED and bulb should remain on.

9. Remove sensor B from water. The green LED and bulb should glow again.

10. Also observe that the blue LED2 glows for two to four seconds every time the green LED turns on.

11. When the green LED is on disconnect sensor D from A'. The LED must go off.

The above tests confirm that the water level sensor, sump sensor and starter activation circuits are functioning properly.

The no-flow delay circuit and auto timer cut-off circuit can be tested using the following procedure:

1. Connect jumper J1 (refer Fig. 2).

2. Select no-flow time, say, 30 seconds, using DIP switch S3, and auto timer cut-off time, say, 30 minutes, using DIP switch S2.

3. Put sensors A' and D into water. The bulb must glow.

4. Remove sensor D from the water. The bulb must go off after 30 seconds (to show that the no-flow delay circuit is functioning properly).